

Precision-Wormhole: Bridging Reach and Stability in High-Stakes VR Remote Manipulation

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ABSTRACT

High-stakes VR remote manipulation requires balancing extensive reach and meticulous precision, yet current paradigms struggle with tremor amplification or depth ambiguity. We introduce *Precision-Wormhole*, a hybrid technique augmenting standard Wormhole with Context-Aware Motion Damping to filter jitter and a Proximity-Triggered Visual Magnifier to recover depth cues. Our evaluation confirms that *Precision-Wormhole* significantly outperforms Go-Go in success rate while maintaining the low workload and high embodiment of the 1:1 baseline. We identify a stability-precision trade-off: magnification enhances accuracy at the cost of increased clutching, indicating that visual clarity shifts user strategy from “satisficing” to “optimizing.”

Index Terms: Virtual Reality, Remote Manipulation, 3D Interaction, Hybrid Techniques, Precision-Wormhole.

1 INTRODUCTION

Interacting with objects beyond physical reach is a critical requirement for high-stakes remote applications in Virtual Reality (VR), such as explosive ordnance disposal (EOD), telesurgery, or nuclear decommissioning. In these contexts, operators must maintain a safe distance while executing tasks with meticulous precision, where failure caused by hand tremor or visual misalignment could lead to catastrophic consequences.

However, balancing extensive reach with local stability remains a challenge. As highlighted in foundational evaluations [4], the Go-Go technique (*non-linear arm extension*) amplifies physiological tremors at long ranges. Similarly, while the Wormhole technique (*hand teleportation*) [2] allows direct reach, it often suffers from depth perception issues due to the loss of binocular disparity and can induce arm fatigue.

To address these limitations, we introduce “Precision-Wormhole,” a hybrid technique specifically designed to stabilize distant manipulation. This approach refines the standard Wormhole paradigm by integrating two proximity-based mechanisms: *Context-Aware Motion Damping* to filter motor jitter, and a *Proximity-Triggered Visual Magnifier* to recover critical depth cues.

We present a within-subjects study ($N = 12$) comparing Precision-Wormhole against Go-Go and standard Wormhole. Results confirm our mechanisms effectively bridge the reach-precision gap without compromising workload or embodiment.

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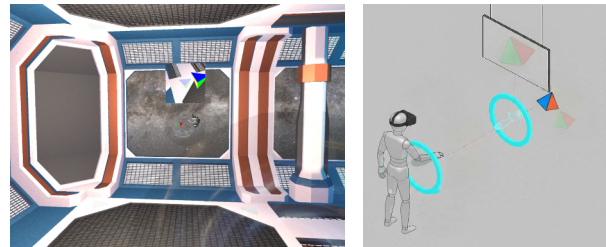


Figure 1: System Overview. (Left) First-person view. (Right) Third-person schematic.

2 SYSTEM DESCRIPTION

The system was developed in Unity 2020.3 and deployed on an HTC Vive Pro HMD using standard controllers. The interaction volume was configured with an entry portal anchored 50 cm in front of the user, mapping to a remote workspace located at a fixed distance of 3.0 m. The target interaction zone was positioned 15 cm beyond this remote exit, resulting in a total functional distance of 3.15 m.

We implemented three distinct interaction techniques. The Go-Go technique served as a baseline for non-linear reach, which extends the user’s reach by applying a non-linear gain to hand movements beyond a threshold. For Go-Go, the threshold (D) was auto-calibrated to 65% of the user’s arm length to allow reaching the target at 90% extension. The standard Wormhole technique, providing a 1:1 remote mapping through a 25 cm portal, allowed users to directly manipulate distant objects as if they were within reach.

Our proposed technique, **Precision-Wormhole**, advances beyond standard Portal-based interactions [5, 1]. While techniques like the Portal widget [5] and Point & Portal [1] facilitate distant interaction, they typically rely on a rigid 1:1 control-display mapping once the hand is inside the portal. In contrast, Precision-Wormhole integrates *Context-Aware Motion Damping*. By leveraging Semantic Pointing [3], we linearly reduce the Control-Display (CD) gain from 1.0 to 0.6 when the virtual hand enters a 5 cm stabilization buffer.

This non-one-to-one mapping distinction is critical: it suppresses position-rotation coupling noise during the docking phase, addressing the stability issues inherent in 1:1 mapping. Additionally, we implemented a Proximity-Triggered Magnifier that provides an optimized “Over-the-Shoulder” perspective. The magnified feed is displayed on a billboard virtual screen floating 40 cm above the wormhole exit, functioning as a 3D Magic Lens [6] to reveal high-frequency spatial details (see Figure 1).

3 STUDY DESIGN

To validate the effectiveness of our proposed technique, we designed a 3×2 within-subjects experiment. Participants performed a 6-DOF object docking task, aiming to align a tetrahedron with a target ghost as accurately and quickly as possible.

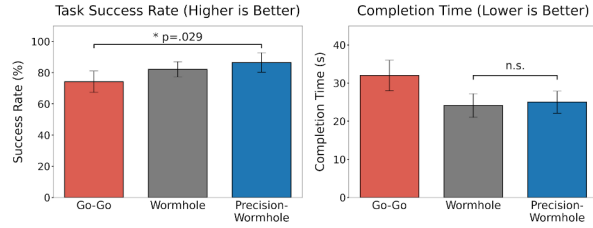


Figure 2: Precision-Wormhole significantly outperforms Go-Go in success rate ($p = .029$) while achieving parity with the 1:1 Wormhole baseline.

3.1 Participants & Experimental Design

Twelve participants (7 male, 5 female; age $M = 23.17$, $SD = 1.75$) were recruited. The experiment employed a within-subjects factorial design: Technique (Go-Go, Wormhole, Precision-Wormhole) $\times$ Magnification (On, Off). To control for order effects, the sequence was counterbalanced using a Latin Square design. In each block, participants completed 2 practice trials followed by 10 measured trials, totaling 720 trials.

3.2 Task & Procedure

Participants performed a high-precision 6-DOF docking task adapted from Ablett et al. [1]. This served as an abstracted micro-task simulating the high-precision alignment phase of remote surgery or EOD tools. The stimulus was a multi-colored tetrahedron (edge length 7 cm). Each trial commenced with a white sphere at a “Home Point.” After a 1.0s dwell time, the sphere disappeared, and a controllable tetrahedron and a semi-transparent ghost target were instantiated at 3.15 m.

Crucially, to simulate realistic manipulation noise and mitigate learning effects, the ghost target was instantiated with **variable offsets of 20° or 45°** along arbitrary axes relative to the initial object orientation.

Participants grasped the tetrahedron and aligned it with the ghost target. Success was confirmed within strict tolerances (Position Error < 2.5 cm, Rotation Error < 8°). A “Skip” mechanism allowed terminating impossible trials.

3.3 Measures

We recorded objective metrics: **Completion Time**, **Success Rate**, **Final Placement Error**, and **Clutching Frequency**. Subjective assessments included the standard NASA-TLX for workload, UEQ-S for user experience, and a custom **Sense of Embodiment (SoE)** questionnaire.

4 RESULTS

Data were analyzed using Repeated Measures ANOVA ($\alpha = .05$) with Bonferroni corrections.

Analysis of **Success Rate** indicated that Precision-Wormhole (PW) significantly outperformed Go-Go ($p = .029$) and matched the standard Wormhole ($p > .99$). Regarding **Completion Time**, results approached significance ($p = .054$), suggesting that PW’s damping mechanics imposed no statistically significant latency compared to the baseline. Furthermore, **Final Position** and **Rotation Error** did not differ significantly across techniques ($p > .05$), confirming that stability aids did not compromise ultimate precision.

Magnification produced a clear trade-off. While it showed a strong trend toward reducing **Final Placement Error** ($p = .057$), it significantly increased **Clutching Frequency** ($p = .013$). Enabling the magnifier improved accuracy but required more re-centering actions.

Regarding subjective metrics, **NASA-TLX** scores revealed that PW imposed the lowest workload ($p = .002$). **UEQ-S** results fur-

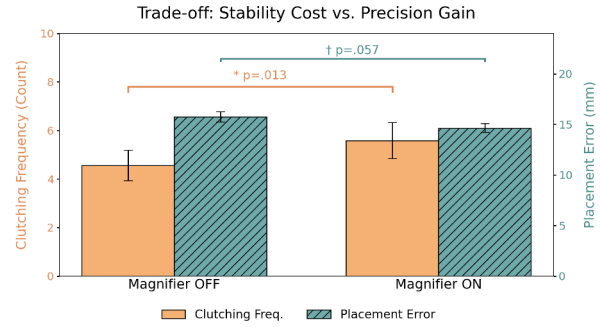


Figure 3: Stability-Precision Trade-off. Magnification shows a trend toward improved accuracy ($p = .057$) but significantly increases clutching frequency ($p = .013$).

ther supported this, showing significant benefits in both Pragmatic ($p = .005$) and Hedonic ($p = .014$) qualities. Crucially, **Sense of Embodiment (SoE)** scores ($p = .005$) indicated that PW elicited a stronger sense of ownership than Go-Go ($p = .03$), contradicting concerns about the disconnect in non-1:1 mapping.

5 DISCUSSION & CONCLUSION

This study proposed and evaluated Precision-Wormhole, an enhanced interaction technique designed to mitigate the practical instability of distant manipulation. A primary contribution lies in demonstrating that PW successfully addresses the limitations of standard 1:1 remote mapping. Proximity-triggered damping effectively creates a stabilization buffer, filtering out the “jitter penalty” while retaining the intuitive 1:1 nature.

Our magnifier findings (Figure 3) suggest enhanced visual clarity shifts user strategy from “satisficing” to “optimizing.” While the magnifier did not impact total task time ($p = .37$), it significantly increased clutching behaviors ($p = .013$) alongside improved accuracy. In domains like cultural heritage restoration or EOD, this trade-off is highly desirable.

In conclusion, Precision-Wormhole proves to be a robust paradigm, delivering high-fidelity precision and a strong sense of embodiment ($p = .005$). We acknowledge the sample size ($N = 12$) as a limitation warranting validation with larger cohorts. Future work will explore adaptive scaling of damping coefficients based on real-time tremor analysis.

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